

A Minimal-Operator-Space Subcase of Local Reflexivity for Completely M -Embedded Spaces

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Status. This is a `partial_result_likely_valid` for a question in Sharma's paper [1]. It gives an affirmative answer for the minimal operator-space structures arising from classical M -embedded Banach spaces. It is not a full answer for arbitrary one-sided M -embedded operator spaces.

Source Question

On page 12, Sharma asks whether right M -embedded or completely M -embedded operator spaces are locally reflexive.

Remark. A natural question is whether right M -embedded or completely M -embedded spaces are locally reflexive? Also note that if X has the completely bounded approximation property then by [ER2, Theorem 11.3.3], X has the strong operator space approximation property. Hence by [ER2, Corollary 11.3.2], X has the slice map property for subspaces of $B(\ell^2)$. There seems some hope that the argument in [ER2, Theorem 14.6.6] can be made to imply that X is 1-exact, and hence is locally reflexive.

Proof Intuition

The source paper already supplies many completely M -embedded examples: if E is a classical M -embedded Banach space, then $\text{Min}(E)$ is completely M -embedded as an operator space. For these examples local reflexivity comes from a different and very robust mechanism: minimal operator spaces sit completely isometrically inside commutative C^* -algebras. Commutative C^* -algebras are nuclear, hence exact, and exactness passes to operator subspaces. Since exact operator spaces are locally reflexive, the minimal examples are locally reflexive without using any special one-sided M -ideal structure beyond Sharma's observation that they are indeed completely M -embedded.

Theorem

Theorem 1. *Let E be an M -embedded Banach space. Then the minimal operator space $\text{Min}(E)$ is completely M -embedded and locally reflexive. Indeed, $\text{Min}(E)$ is 1-exact.*

Proof. Sharma records that if E is an M -embedded Banach space, then $\text{Min}(E)$ is completely M -embedded as an operator space [1, Section 2]. It remains only to prove local reflexivity.

Let $K = \text{Ball}(E^*)$ with its weak-star topology. The canonical Banach-space embedding

$$E \longrightarrow C(K), \quad x \mapsto (\varphi \mapsto \varphi(x)),$$

is isometric. By the defining property of the minimal operator-space structure, this identifies $\text{Min}(E)$ completely isometrically with an operator subspace of the commutative C^* -algebra $C(K)$.

Commutative C^* -algebras are nuclear and therefore 1-exact. Exactness passes to operator subspaces, so $\text{Min}(E)$ is 1-exact. Finally, exact operator spaces are locally reflexive; this is the standard exactness-to-local-reflexivity implication used in operator-space theory and also invoked in the source remark's phrase "1-exact, and hence locally reflexive" [1, p. 12]. Hence $\text{Min}(E)$ is locally reflexive. $\square$

Verification and Novelty Notes

The proof is a chain of standard implications and contains no computational component. The verification focus is whether one accepts the standard operator-space facts: minimal spaces embed into commutative C^* -algebras, commutative C^* -algebras are 1-exact, exactness passes to subspaces, and exact operator spaces are locally reflexive.

The cheap indexes were searched for arXiv:0902.4257, one-sided M -embedded spaces, completely M -embedded spaces, local reflexivity, exactness, and $C_\infty(X)$. Bounded web searches for exact local-reflexivity phrases found the source paper but no separate full resolution. Since the proof uses standard operator-space facts, novelty confidence is low-to-moderate; the packet is best viewed as a useful explicit subcase and duplicate-memory entry.

Limitations

This does not answer whether all right M -embedded or completely M -embedded operator spaces are locally reflexive. It only treats the minimal operator-space structures $\text{Min}(E)$ coming from classical M -embedded Banach spaces. In particular, it does not address the non-minimal and genuinely one-sided nonselfadjoint examples emphasized in [1].

Human Review Recommendation

Reviewers should check that the packet's operator-space background facts are being used in their standard completely isometric form. The mathematical claim is intentionally modest; the main value is explicitly connecting the source question with the minimal examples.

References

- [1] S. Sharma, *Operator Spaces which are One-sided M -Ideals in their bidual*, arXiv:0902.4257, 2009.